Thesis Template

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Abstract

Cybersecurity threats in the energy sector are becoming increasingly nu- merous and sophisticated. As Operational Technology (OT) environments become more interconnected, organisations must process growing volumes of cyber threat intelligence. The manual analysis of this information is time consuming and expensive, creating a need for automation of methods which can support threat assessment and decision making. The objective of this thesis is to investigate the use of RAG-enhanced LLMs for cyber threat analysis in the energy sector. The method to be used will involve collecting cyber threat intelligence sources relevant to the energy sector. This includes CVEs, security advisories and threat reports. The second stage will be to build a RAG pipeline using an existing LLM. Comparisons will be made between using the LLM alone and the RAG-enhanced LLM. Finally, evaluate the quality of the generated threat analysis. The thesis is expected to provide an assessment of the value of Retrieval- Augmented Generation in cyber threat analysis in the energy sector. Addi- tionally, the thesis aims to identify potential benefits and limitations of using RAG-enhanced LLMs in cyber threat intelligence tasks.

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Chapter 1

Introduction

Operational Technology (OT) is made of programmable systems and devices that interact with the physical environment [1].

Critical National Infrastructure (CNI) is defined as services and infrastructure which are essential to the health, safety, security or well-being of citizens. This includes services such as water purification plants, transportation and energy production. All of these CNIs make use of network technologies, involving IoT devices, which allow for data collection, communication and analysis [2].

OT devices are increasingly connected in order to reap the benefits of remote monitoring and predictive analytics. However, this increased connectivity involves the risk of cyber attacks. The results of attacks may include downtime, safety incidents or environmental damage [3].

A possible reason for existing security threats is that OT environments were not designed in accordance with modern connectivity. The lack of security features can be difficult to solve. Additionally, the specific risks need to be identified along with the respective threats [3].

Common Vulnerability Exposures (CVE) list computer security flaws and are made available to the public. Each CVE receives a CVE ID number. CVE IDs provide users with a reliable method of identifying vulnerabilities and to advance security and solutions. Security advisories will almost always refer to CVEs. The information from CVEs assists in the efforts of professionals to improve security in systems such as OT networks [4]. Other sources of information which security advisories will refer to include books, blogs, research papers and news [5].

Manually sifting through threat reports and CVEs would be labour intensive, expensive and inefficient. Large Language Models (LLMs) have been rapidly advancing and are opening opportunities for increased cyber security. Cyber threats continue to grow and therefore, the need for automatic detection and analysis of vulnerabilities become more pressing. In the world of network security, LLMs have proven to be effective in detecting and classifying attacks from network traffic data as well as cyber threat analysis [5].

The use of LLMs in Network Threat Analysis is categorised into Cyber Threat Intelligence (CTI) generation and CTI analysis for decision making. CTI reporting has become essential in modern risk management. As the volume of CTI reports

increases, so grows the need for automated CTI Analysis. LLMs have proven the ability to generate reports for potential cybersecurity threats and predict their impact [5].

While LLMs on their own are useful, they are limited by the data available during training. In the field of cyber security, information changes rapidly therefore, access to current knowledge is important. Enhancing LLMs with Retrieval Augmented Generation (RAG) in a niche area, such as cyber security in the energy sector, can be a powerful tool. RAG addresses the limitations of LLMs by using a retriever to source relevant information. This information is then appended to the query and passed to the LLM [6].

1.1 Research Questions

Research questions to be explored in this thesis report are as follows:

1. How can Retrieval-Augmented Generation be implemented to assist in cyber threat intelligence analysis in the energy sector?

2. How do industry professionals perceive the use of RAG-enhanced LLMs in cybersecurity intelligence analysis?

Chapter 2

Background

2.1 Cyber Threat Intelligence in OT and Energy Systems

Modern OT came about from the addition of IT systems into existing infrastructure, mostly to replace or improve physical control systems. This evolution has been pushed by improvements in cost and performance, which have in turn resulted in modern smart technologies, making up the Internet of Things. The increase in efficiency and optimisation of these systems are positive, however the connectivity opens them up to potential attack. A major design consideration in OT systems, according to NIST, is the ability to detect unsafe conditions and trigger solutions to resolve the problem. This in turn creates a need for adaptability, resilience, safety and security. According to NIST, OT can be found in all critical infrastructure, especially the energy sector, which this thesis report focuses on. [1]

NIST describes a threat as "any circumstance or event with the potential to adversely impact organisational operations, assets, individuals, other organisations, or the nation through an information system via unauthorised access, destruction, disclosure, modification of information, and/or denial of service. An incident is defined as when a threat event actually occurs and causes damage or leaks sensitive information. Additionally, NIST defines a vulnerability as "a weakness in an information system (including an OT), system security procedures, internal controls, or implementation that could be exploited or triggered by a threat source." Finally, the intent or method which exploits a vulnerability is known as the threat source. [1]

In the United States the Cybersecurity and Infrastructure Security Agency provides warnings to CTI owners and operators regarding potential threats to networks. Additionally, MITRE ATTCK for Industrial Control Systems (ICS) provides a database for adversarial activity in the ICS field. This emphasises the necessity for cybersecurity within Operational Technology. [1]

Supervisory Control and Data Acquisition (SCADA) technology is used in the energy sector to operate interconnected systems made up of private and public utilities. Certain SCADA systems monitor and control infrastructure by gathering data and sending commands control stations. Distributed Control Systems (DCS) and SCADA often work together. DCS will control a power generation facility, but must coordinate with the SCADA system in order to manage output and distribution demands. [1]

Operational Technology (OT) includes a wide range of systems and devices that interact with the physical environment. The various systems are responsible for changes and detection via controlling and monitoring devices, processes and events. NIST lists examples such as industrial control systems (ICS), building automation systems, transportation systems, physical access control systems, monitoring systems and measurement systems. OT is an important part of critical national infrastructure (CNI). These CNIs have become less isolated and more connected via IoT devices as technology has progressed. This connectivity puts the OT systems at risk of adversarial attacks. Security solutions for Information Technology (IT) systems exist and are well established. However, security solutions for OT systems need to be implemented with caution as they have specific requirements. One of the fundamental differences between IT and OT systems is that OT systems have a direct effect on the physical world. These systems have the potential to affect the health and safety of citizens, environmental damage, financial loss and leakage of sensitive information. [1]

NIST explains that risk assessments require identifying vulnerabilities and threats within an OT system. It is necessary to keep an inventory of the IT and OT devices in the system, which assists in identifying, tracking and solving vulnerabilities.

2.2 Existing Methods for Cyber Threat Intelligence Analysis

Cyber Threat Intelligence (CTI) is a collection of evidence of information received from various sources with a focus on actionable response towards threats. Additionally, the data assists in monitoring and keeping up to date on threat profiles. Threat profiles are a source of information on threat campaigns, victims, motivation, tools, and methodologies which can be used to attack networks. [7]

Structured Threat Information Expression (STIX) is an effort to create a standardised language which defines cyber threats and documents their instances. The objectiev of STIX is to share threat indicators, information management, exploitation of vulnerabilities, tactics, techniques, incident location, and actions to be taken. STIX is shared among to trusted actors using Trusted Automated eXchange of Indicator Information (TAXII), providing awareness of of emerging threats in the hope that they will be detected and stopped in time. [7]

Qamar et al states that the information on CTI from multiple sources is essential to cybersecuroty and that it needs to be normalised and validated. An example is given from the finance industry where Financial Services Information Sharing and Analysis Center regularly maintain and share CTI data for players in the world of finance. Additionally, Hail a Taxii cooperates with various communities in order to provide free CTI data. Qamar adds that considering the amount and complexity of data collected, manual analytics are not feasible. [7]

While the communities mentioned above share valuable information and have good intent, the method is not without issues. For example, STIX reports are manually generated by security analysts. Validation of these reports is not always possible before they are shared. This leads to CTI data which can be incomplete or just incorrect. Additionally, multiple analysts often report a single threat, which leads to many opposing interpretations of that one threat. Such inconsistencies make it

difficult to accurately analyse reported threats. [7]

Qamar et al explains that before CTI standards were in place, experts would report their analyses on web blogs. Latent Semantic Analysis (LSA) was then used to find semantically relevant information in the reported findings. Key words from the various topics are given quantitative importance with the use of Probablistic LSA (PLSA). This method is limited and outdated due to how broad the search is in comparison to the volume of information from web blogs.

Qamar et al. [7] proposed STIX-Analyzer, a framework for automated cyber threat intelligence analysis using ontologies and semantic reasoning. The framework takes structured threat intelligence using STIX with vulnerability information from CVEs and network configuration data. This information is represented using the Web Ontology Language (OWL), which enables relationships between threats, vulnerabilities and network components to be formally defined.

The framework uses logical inference rules expressed in the Semantic Web Rule Language (SWRL) to analyse these relationships. This allows the system to identify threats relevant to a particular network, determine potentially affected assets and assess associated risks. Qamar demonstrates an approach to automated CTI analysis in which structured threat information is combined with knowledge of the target environment to support decision making in cybersecurity.

However, the approach depends on the quality and completeness of the supplied threat intelligence and network information, which the authors identify as a limitation.

2.3 Large Language Models in Threat Detection

Several different Large Language Models (LLMs) are available to industry and the public. These LLMs have shown that they can be valuable tools in various fields such as finance, healthcare and education. Word2Vec is a method that maps words which are semantically similar to nearby positions in a vector space. This allows the model to make predictions based on context rather than basic similarity. While LLMs are powerful tool, they do have their drawbacks. Chen et al lists issues such as prompt injection, data poisoning attacks and backdoor attacks which can compromise the quality of output generated by LLMs. Chen continues by mentioning that the cost of high-performance computing needs to be taken into account, as this can prevent deployment on local systems at a lower cost. [8]

The internet, since its inception, has been faced with issues concerning vulnerabilities and attacks. To add to the complexity of an already multifaceted problem, the Internet of Things (IoT) has increased connectivity and made threats even less predictable.

2.4 Retrieval-Augmented Generation

Lewis et al ([6]) compare models using parametric and non-parametric memory. Parametric memory is described as information which is baked into the model during training. This information is readily available, but the world changes and information can be incomplete. Updates to the model's parameters can be inconvenient and difficult as it may involve retraining. Additionally, the model does not refer to the source of specific facts and can generate answers that seem plausible, but are

incorrect. This is because the model predicts the probability of the next token given the previous tokens.

RAG offers a solution in the form of an external source of information while the model generates an answer. Lewis et al refer to the external database as non-parametric memory. Therefore, the model will not only rely on what it learned during training, but rather it first retrieves relevant passages as context for generation. Lewis et al ([6]) note that an important advantage is that the external database can be updated without having to retrain the whole language model. It is also easier to view the source of the information generated by the model, which allows for moderation. These advantages make RAG especially attractive for domains, such as cybersecurity, where updates occur frequently and those updates are important to maintaining a secure system.

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Chapter 3

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public class HelloWorld<T> extends Something<T> {
  public static void main(String[] args) {
    System.out.println("Hello, world!");
  }
}
```

You can also get it with line numbers:

```
1 public class HelloWorld<T> extends Something<T> {
2   public static void main(String[] args) {
3     System.out.println("Hello, world!");
4   }
5 }
```

You can refer to lines like this: `\cref{code:hello}` (gives you Chapter 3, see source for more details).

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  public static void main(String[] args) {
    System.out.println("Hello, world!");
  }
}
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Chapter 4

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Chapter 5

Implementation

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Chapter 6

Evaluation

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Chapter 7

Conclusions

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